

Lecture 9: Linear Programming

FEBRUARY 16

Lecturer: Mitchell Scott

Disclaimer: *These notes have not been subjected to the usual scrutiny reserved for formal publications. They may be distributed outside this class only with the permission of the Instructor.*

9.1 Linear Programming

Remark 9.1 *I said the very first day that game theory was all about optimization. Now that's what we are going to be doing.*

Example 9.2 (Lumber Yard) *A lumber mill saw finish grade and construction grade boards from the logs that it receives. Suppose that it takes two hours to rough saw each one thousand board feet of the finish grade boards and five hours to plan each 1000 board feet of these boards suppose also that it takes two hours too rough saw each 1000 feet of construction grade boards but it takes only three hours to pay in each 1000 board feet of these boards. The size is available eight hours per day and the plane is available 15 hours per day. If the profit on each 1000 board feet of finish grade boards is \$120 and the profit on each 1000 board feet of construction grade board is \$100, how many board fee of each type of lumber should be sold to maximize profit?*

Mitchell Scott: *A lumber mill saw finish grade and construction grade boards from the logs that it receives. Suppose that it takes two hours to rough saw each one thousand board feet of the finish grade boards and five hours to plan each 1000 board feet of these boards suppose also that it takes two hours too rough saw each 1004 feet of construction grade boards but it takes only three hours to pay in each 1000 board feet of these boards. The size is available eight hours per day and the plane is available 15 hours per day. If the profit on each 1000 board feet of finish grade boards is \$120 and the profit on each 1000 board fee of construction grade board is \$100, how many board fee of each type of lumber should be sold to maximize profit?*

Let f and c be the amount of finish-grade boards and construction-grade boards, respectively, measured in 1000 board feet.

For the saw is only available 8 hours, we have

$$2f + 2c \leq 8.$$

For the plane, which is available 15 hours, we have

$$5f + 3c \leq 15.$$

Of course, we have $f \geq 0, c \geq 0$. The profit (in dollars) to be maximized is

$$z = 120f + 100c.$$

where only one $\geq, \leq$ or $=$ appears. We call the linear function 9.1 the objective function. The (in)equalities in 9.2 are called the constraints.

Definition 9.6 (Standard form) We will say that the standard form of the linear programming problem is it has the form: Find values of $x_1, x_2, \dots, x_n$ that will

$$\text{minimize } z = c_1x_1 + c_2x_2 + \dots + c_nx_n \text{ subject to the restrictions} \quad (9.3)$$

$$\left. \begin{array}{l} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \leq b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n \leq b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n \leq b_m \end{array} \right\} \quad (9.4)$$

$$x_j \geq 0, \quad \text{for all } j = 1, 2, \dots, n \quad (9.5)$$

Definition 9.7 (Canonical form) We will say that the canonical form of the linear programming problem is it has the form: Find values of $x_1, x_2, \dots, x_n$ that will

$$\text{minimize } z = c_1x_1 + c_2x_2 + \dots + c_nx_n \text{ subject to the restrictions}$$

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

$$x_j \geq 0, \quad \text{for all } j = 1, 2, \dots, n$$

Example 9.8 (A canonical form example) Find values of x, y, u and v that will

$$\text{minimize } z = -3x - 2y - 3u + 4v \text{ subject to the restrictions}$$

$$2x + y + 2u - v = 4$$

$$5x + 3y - 2v = 15$$

$$x \geq 0, y \geq 0, u \geq 0, v \geq 0,$$

9.2 Getting into Canonical form

Every minimization problem can be viewed as a maximization problem.

$$\min c_1x_1 + c_2x_2 + \dots + c_nx_n = \min \sum_{i=1}^n c_i x_i = -\max \left(-\sum_{i=1}^n c_i x_i \right) = -\max -c_1x_1 - c_2x_2 - \dots - c_nx_n$$

If we multiply an inequality

$$c_1x_1 + c_2x_2 + \dots + c_nx_n \geq b$$

by -1, we get

$$-c_1x_1 - c_2x_2 - \dots - c_nx_n \leq -b$$

To change an equality to an inequality we need to observe that $x = 6$ is the same as $x \leq 6$ and $x \geq 6$, or $-x \leq -6$. For a generic pair of inequalities,

$$\sum_{j=1}^n a_{ij}x_j = b_i,$$

we could get to

$$\begin{aligned}\sum_{j=1}^n a_{ij}x_j &\leq b_i \\ \sum_{j=1}^n -a_{ij}x_j &\leq -b_i\end{aligned}$$

While we want all variables to be ≥ 0 , that is not always the case, so assume x_j is not constrained to be nonnegative. We can constrain it to be $x_j = x_j^+ - x_j^-$, where $x_j^+, x_j^- \geq 0$. That is any number can be written as the difference of two nonnegative numbers.

Lastly as we talked about with linearity, if we scale one side of the constraints of a linear programming problem, then the optimal solution is the same as the adjusted problem.

Example 9.9 (A Blending Problem) *A manufacturer of artificial sweetener blends 14 kg of saccharine and 18 kg of dextrose to prepare two new products: SWEET and LO-SUGAR. Each kilogram of SWEET contains 0.4 kg of dextrose and 0.2 kg of saccharin, whereas each kilogram of LO-SUGAR contains 0.3 kg of dextrose and 0.4 kg of saccharin. If the profit on each kilogram of SWEET is \$0.20 and the profit on each kilogram of LO-SUGAR is \$0.30 how many kilograms of each product should be made to maximize the profit?*

Let x, y be the number of kg of SWEET and LO-SUGAR made respectively.

To satisfy the dextrose we have

$$0.4x + 0.3y \leq 18.$$

To satisfy the saccharin, we have

$$0.2x + 0.4y \leq 14.$$

Obviously $x, y \geq 0$ and the total profit we seek to maximize is $z = 0.20x + 0.30y$.

Our model is

Find values of x, y that will

$$\begin{aligned}\text{maximize } z &= 0.20x + 0.30y \text{ subject to the restrictions} \\ 0.4x + 0.3y &\leq 18 \\ 0.2x + 0.4y &\leq 14 \\ x &\geq 0 \\ y &\geq 0\end{aligned}$$

Example 9.10 (A Financial Problem) *Let x and y denote the amounts invested in bond AAA and stock*

BB , respectively. We then have

$$\begin{aligned}x + y &= 100,000 \\x &\geq 2y \\y &\leq 30,000 \\x &\geq 0 \\y &\geq 0\end{aligned}$$

That means the return to the university which is to maximize is $z = 0.07x + 0.09y$.

That means Find values of x, y that will

$$\begin{aligned}\text{maximize } z &= 0.07x + 0.09y \text{ subject to the restrictions} \\x + y &= 100,000 \\x - 2y &\geq 0 \\y &\leq 30,000 \\x &\geq 0 \\y &\geq 0\end{aligned}$$

9.3 Matrix Equation

Consider the standard form of the linear programming problem:

Find values of $x_1, x_2, \dots, x_n$ that will

$$\text{minimize } z = c_1x_1 + c_2x_2 + \dots + c_nx_n \text{ subject to the restrictions} \quad (9.6)$$

$$\left. \begin{aligned}a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &\leq b_1 \\a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &\leq b_2 \\&\vdots \\a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &\leq b_m\end{aligned} \right\} \quad (9.7)$$

$$x_j \geq 0, \quad \text{for all } j = 1, 2, \dots, n \quad (9.8)$$

If we let

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}, \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}, \mathbf{c} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix}$$

We can write this as Find the vector $\mathbf{x} \in \mathbb{R}^n$ that will

$$\text{minimize } z = \mathbf{c}^\top \mathbf{x} \text{ subject to the restrictions} \quad (9.9)$$

$$\left. \begin{aligned}\mathbf{Ax} &\leq \mathbf{b} \\ \mathbf{x} &\geq \mathbf{0}\end{aligned} \right\} \quad (9.10)$$

$$(9.11)$$